which leads to the probability density

$$\rho_{\mathrm{NPT}} = Z_{\mathrm{NPT}}^{-1} e^{-\beta(H+\mathrm{PV})},$$

where H is the Hamiltonian, and $\beta^{-1} = \mathrm{k_B T}$.

The NPT model may be viewed as corresponding to a thermostatted system which incorporates a "piston" controlling the overall volume of the system to be consistent with the target pressure. During simulation, the volume must be free to change; whereas the methods presented so far retain a fixed volume. In NVE or NVT simulation, the volume enters in two ways into the molecular simulation method: (i) as a modification of the potential (allowing for periodic replicas) and (ii) in the mechanism by which atoms are mapped from one boundary to the other of the rectangular domain, simulating motion on a torus. It is not obvious how to generalize these to the NPT setting where V is a dynamical variable.

The method of Andersen [11] ("Andersen's piston") is to control the volume by reparamaterizing the coordinates to lie in a cubic box with unit side. Assuming the system occupies a cubic domain, say, $[-L/2, L/2] \times [-L/2, L/2] \times [-L/2, L/2]$, where $L = V^{1/3}$, we may write, for each three-dimensional atomic coordinate vector $\boldsymbol{q}_i$

$$\boldsymbol{q}_i = V^{1/3} \bar{\boldsymbol{q}}_i, \tag{8.71}$$

where $\bar{\boldsymbol{q}}_i$ is confined to the unit-side cube. Then we have

$$\dot{\boldsymbol{q}}_i = \frac{1}{3} V^{-2/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{1/3} \frac{\mathrm{d}}{\mathrm{d}t} \bar{\boldsymbol{q}}_i.$$

The kinetic energy of the atomic system may be written

$$K = \sum_{i=1}^{N} m_i \frac{\|\dot{\boldsymbol{q}}_i\|^2}{2}.$$

After coordinate transformation this becomes

$$\begin{aligned} \bar{K} &= \sum_{i=1}^{N} \frac{m_i}{2} \left\| \frac{1}{3} V^{-2/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{1/3} \frac{\mathrm{d}}{\mathrm{d}t} \bar{\boldsymbol{q}}_i \right\|^2 \\ &= \sum_{i=1}^{N} \frac{m_i}{2} \left[\frac{1}{9} V^{-4/3} \dot{V}^2 \left\| \bar{\boldsymbol{q}}_i \right\|^2 + \frac{2}{3} V^{-1/3} \dot{V} \bar{\boldsymbol{q}}_i \cdot \dot{\bar{\boldsymbol{q}}}_i + V^{2/3} \left\| \dot{\bar{\boldsymbol{q}}}_i \right\|^2 \right], \end{aligned}$$

thus the momentum vector conjugate to $\bar{\boldsymbol{q}}_i$ becomes

$$\bar{\boldsymbol{p}}_i = \frac{\partial \bar{K}}{\partial \dot{\bar{\boldsymbol{q}}}_i}$$

$$= m_i \left[\frac{1}{3} V^{-1/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{2/3} \dot{\bar{\boldsymbol{q}}}_i \right],$$

from which, we observe

$$\bar{\boldsymbol{p}}_i = m_i V^{1/3} \dot{\boldsymbol{q}}_i. \tag{8.72}$$

There is still the question of how to adjust the volume in simulation. The idea of Andersen was to view the volume V as a dynamical quantity on equal footing with the coordinates of the system by viewing PV as the potential energy term and introducing an artificial kinetic energy term of the form $\mu_V \dot{V}^2$. The barostat mass μ_V becomes a free parameter, analogous to the thermal mass used in Nosé dynamics. An extended Hamiltonian may then be written by introducing a momentum variable p_V conjugate to V:

$$\hat{H}(\bar{\boldsymbol{q}}, \bar{\boldsymbol{p}}, V, p_V) = \sum_{i=1}^{N} \frac{\|\bar{\boldsymbol{p}}_i\|^2}{2m_i V^{2/3}} + \bar{U}(\bar{\boldsymbol{q}}; V) + \frac{p_V^2}{2\mu_V} + \mathrm{P}V, \tag{8.73}$$

where we have written $\bar{U}(\bar{\boldsymbol{q}}; V) = U(V^{1/3}\bar{\boldsymbol{q}})$.

In the extended space, we then observe that

$$\rho_{\mathrm{NPT}} = \sqrt{\frac{\beta}{2\pi\mu_V}} \int e^{-\beta \hat{H}(\bar{\boldsymbol{q}}, \bar{\boldsymbol{p}}, V, p_V)} \mathrm{d}p_V,$$

from the definitions (8.71) and (8.72).

There are several ways to proceed. Andersen suggested to use a Monte-Carlo method to control the temperature during simulation. This approach has the advantage of simplicity, but only allows the determination of configurational averages. If dynamical information is required (e.g. an autocorrelation function or a diffusion coefficient), some sort of dynamics-based scheme is needed. The progress in developing formulations and numerical methods can be charted in [11, 32, 128, 183, 199, 248, 252, 288, 373].

Deterministic methods can be used based on Nosé-Hoover to control the temperature. A problem with methods of this type that one sometimes encounters "ringing" or "breathing" depending on the tuning of various parameters.

Stochastic-dynamical thermostats greatly improve control of the convergence to equilibrium. We have already seen several methods of this type, specifically the Langevin dynamics method and the Nosé-Hoover-Langevin method. These schemes may be applied in conjunction with the Andersen Hamiltonian $\hat{H}$. The extended system approach outlined here was developed in collaboration with R. Elber and M. Di Pierro; it is close in spirit to the method of Klein [199], but incorporates stochastic terms as in Nosé-Hoover-Langevin in the equations defining the auxiliary variables to ensure ergodicity.